

Attention-Based Phonetic Convolutional Recurrent Neural Networks for Language Identification

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Overview

- 1 Introduction
- 2 Motivation
- 3 Methodology
- 4 Experimental Results
- 5 Conclusions

Introduction

- Language identification(LID) refers to identifying the language of a given spoken utterance.
- It plays a vital role in downstream tasks like the multi-lingual ASR system and speech to speech translations.
- Building the LID systems for the Indian languages is challenging due to common speech patterns across the languages.
- LID can be performed in two ways, namely, acoustic and phonotactic methods.
- In Acoustic methods, short-term spectral feature extraction followed by modeling of LID system.
- In phonotactic methods, we transcribe the speech signal to a sequence of phonemes and use n-gram language models on the obtained phoneme sequence.

Motivation

- Exploiting the spectral information is not sufficient to distinguish the languages.
- The language cues like phonemes and their co-occurrences are an important component while distinguishing the languages.
- The phonetic feature-based methods shown improvement over the acoustic methods.
- Transcribing the speech data requires a large amount of time and patience, and linguistic knowledge.
- We investigate the usage of pre-trained rich resource phonetic discriminator to extract the phonetic features for low resource target languages.

Methodology

- The overall architecture consists of 2 stages
 - Discriminative phonetic feature extraction
 - Training of the LID system

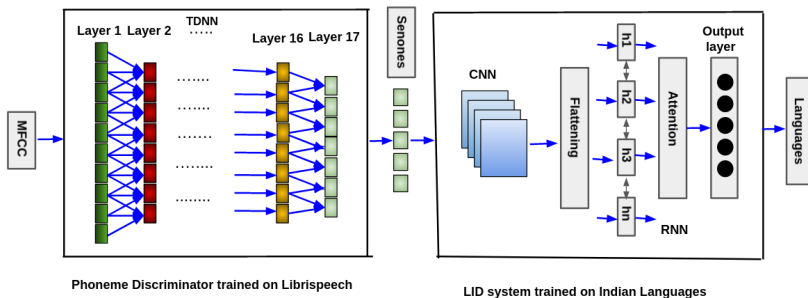


Figure: Proposed architecture for Language Identification

Discriminative phonetic feature extraction

- Training the phoneme discriminator plays an important role.
- We took the pre-trained Librispeech acoustic model for extracting the phonetic features.
- The phoneme discriminator network is a time-delay neural network (TDNN) with sequence discriminative training criterion.
- The pre-final layer is considered as the bottleneck layer for the phonetic feature extraction.

Training of the LID system

- The phonetic features have been extracted from the bottleneck layer.
- These higher-dimensional features are projected to lower-dimensional space using the Affine transform.
- The projected features are fed to three different neural networks to obtain high-level frame representations.
- Attention mechanism is used to pool the high-level frame representations.
- Finally the utterance level representation is classified using softmax.

Network Description & Hyper parameters

- The input features for the TDNN network are 40-dimensional high-resolution MFCC's along with 100-dimensional i-vectors for speaker adaption
- The phoneme discriminator consists of 17 factorized TDNN layers.
- The input to network are fixed to 3 seconds.
- We used two 2-D convolution layers of 32 kernels with (10,10), (3,3) shapes, respectively and a max-pooling layer with the shape of (2,2).
- 128-dimensional BLSTM layer is used followed by multi-head attention.
- We used L2 regularization and 20% dropout for recurrent layers to avoid overfitting.

Dataset Description

- We have used 960 hours of Librispeech data to pre-train the senone classifier.
- LID dataset containing speech data from 7 Indian languages, viz., Hindi, Bengali, Punjabi, Malayalam, Marathi, Tamil, and Telugu.
- The dataset has been collected from a Youtube channel named Josh talks.
- This channel contains motivational, business and real-life story talks from several speakers.
- The trainset for each language consists of 9 hours from 80 speakers and the testset consists of 1.5 hours from 12 speakers.

Table: Performance in different feature settings

Method	Front End	Pooling method	Accuracy
Acoustic-BLSTM	BLSTM	Attention	58.42
Acoustic-Resnet-BLSTM	Resnet-BLSTM	Attention	67.09
Acoustic-CRNN	CRNN	Attention	70.69
Phonetic-BLSTM	BLSTM	Attention	89.13
Phonetic-Resnet-BLSTM	Resnet-BLSTM	Attention	90.1
Phonetic-CRNN	CRNN	Attention	90.73

Table: Duration wise analysis for attention based Phonetic-CRNN system

System	Duration range(s)	Utterances count	Accuracy
Phonetic-CRNN	2-3	2796	85.05
	3-4	2200	90.72
	4-5	1457	93.20
	5-6	861	93.84
	6-10	1184	96.45
	10-30	340	99.11

Learned language embeddings visualization

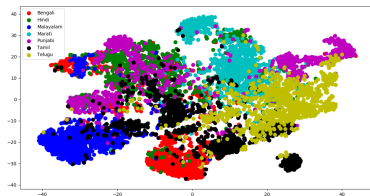


Figure: t-SNE visualization of test data embeddings extracted from the Acoustic-CRNN model.

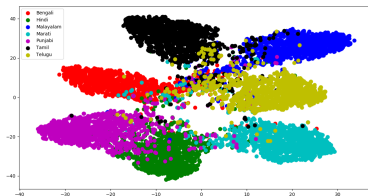


Figure: t-SNE visualization of test data embeddings extracted from the Phonetic-CRNN model.

Confusion matrix for Phonetic-CRNN system

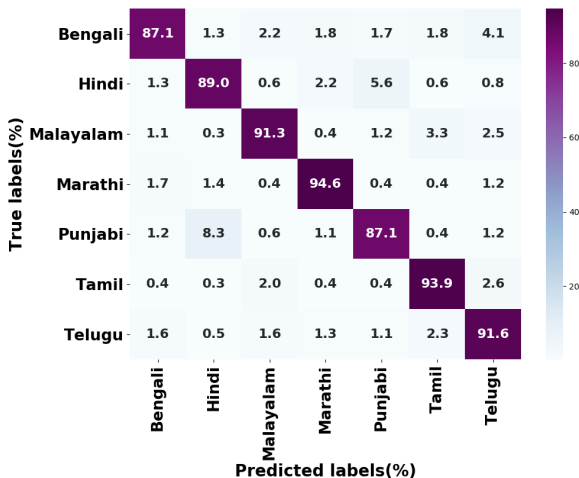


Figure: Confusion matrix for the attention-based Phonetic-CRNN system

Conclusions

- We have observed a 20% absolute improvement for Attention-based Phonetic-CRNN models over the Acoustic-CRNN models.
- The phonetic features have been extracted without knowing the target language transcriptions.
- Language-independent phonetic features play a vital role in low-resource LID systems.
- Working on building LID system for all the 23 official languages.

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Thank you .. Any Queries